

PQ-DID: fixed-pad extraction and encrypted-opening composition

Construction analysis and implementation handover

1 October 2026

Research derivations for a specified ideal-oracle candidate. Not a security certification, adopted profile or native implementation.

Decision

There is concrete progress beyond the previous list of obligations. This note defines an efficient pad decoder, derives component round-by-round binding bounds, gives an explicit conditional knowledge-composition proof and a complete-view simulation argument in an ideal-oracle model. It also gives an algebraic counterexample to sampling only a message-length prefix of an encoded Hadamard-check oracle.

It does not establish complete PQ-DID security or admit the pinned native backend. The full native base-protocol extraction argument, exhaustive disclosure correspondence, exact quantum compiler instantiation and finite parameters are still missing. The claims below are mathematical derivations for the explicitly stated candidate and premises; they are not claims that the cited papers have already proved this exact composition.

Requested obligation	Result of this analysis
Fixed recoverable pad	A total, polynomial-time ideal-oracle decoder is specified. Near and far commitments have separate treatments; a default value is not silently treated as a valid opening.
Knowledge preservation	Component binding states and a wrapper composition lemma are derived. The complete conclusion remains conditional on object-consistent round-by-round knowledge of the chosen base protocol.
Complete privacy/compiler	An ideal-view simulator is specified under explicit independence and disclosure conditions. Native coverage and the quantum/compiler application remain open.

Scope. Only manuscript Sections II–VIII and agreed clarifications determine the authentication relation. The ML-DSA KYC baseline, historical observations, active profile, manuscript, WSL ledgers and paused proof/isolation work have not been changed. No native tests, builds, proof generation or implementation benchmarks were performed here.

1 Chosen interface and evidence

VEIL Definition 3.1 fixes a bound witness before quantifying over subsequent statements. Its Lemma 3.9 supplies a candidate circuit-evaluation interface; Lemma 4.3 supplies a zero-knowledge/soundness compiler. These do not, by citation alone, supply our required round-by-round knowledge theorem [1]. The quantum BCS result requires that stronger knowledge property of the underlying IOP [2].

The local evidence is the retained Binius64 revision 441fbf51ff0bcb0bcd28f3f1b73f4954029e8577, the completed embedding report, the committed-view report and the previously supplied wrapper candidate. Those sources establish that the examined native ZK path already combines inner reductions, an outer protocol and polynomial openings. They do not establish the new construction below.

The intended change is to start from the *unwrapped* base authentication proof, protect its original oracle encodings, and encrypt its sensitive arithmetic transcript through the end of the algebraic opening verifier. A separate complete outer protocol checks the decrypted acceptance predicate. The old outer precommit/private/Libra objects must not accidentally be retained as an additional recursive layer.

The two IntMul claims use one logical object Y with two operands; the trace has its own object. This remains a same-object condition inside the base verifier. It is not implemented by adding independently asserted host-side names or by calling an API “ZK”.

2 A total fixed-pad decoder

Work over a finite field $\mathbb{F}$ of cardinality Q . Fix before commitment all code parameters, symbol representations, message dimensions, layout identifiers and payload projections. Let $E : \mathbb{F}^K \rightarrow \mathbb{F}^N$ be a Reed–Solomon evaluation code, composed, if needed, with a fixed efficiently invertible change of message basis. Its relative minimum distance is $\Delta = (N - K + 1)/N$. Distance between interleaved words counts a position as different if *any* component differs.

An honest pad commitment has two components

$$H = (E(u), E(v)), \quad u = (h, d, p),$$

where $h \in \mathbb{F}^s$ is the pad, d contains any specified auxiliary entries, p is independent query-hiding padding, and v is an independent full companion message. The payload projection selecting h is fixed by the layout. The decoder below works on the complete ideal oracle string H , not on its Merkle root.

Choose $0 < \delta < \Delta/2$ and set $t = \lceil \delta N \rceil - 1$. Define $\text{Dec}(H)$ as follows:

1. Bounded-distance decode each component of H to radius t . Reject malformed shapes or field encodings.
2. Re-encode the two recovered messages and count the union of their disagreement positions with H .
3. If both decodings exist and that union has size at most t , return (near, u, v, h) .
4. Otherwise return $(\text{far}, \perp, \perp, 0^s)$.

Write $d(H)$ for the final pad component of this output. No global nearest-codeword search or arbitrary tie-breaking is used. The outer evaluator may also use the auxiliary entries of the same recovered u ; the pad extractor always projects its fixed pad positions. These are not independently chosen decodings.

Proposition 1 (Fixed efficient decoding). *For this code family and radius, d is a total polynomial-time function of the commitment oracle alone. If H lies at interleaved distance less than δ from a pair of codewords, the near branch recovers the unique pair, hence the same pad for every subsequent statement. Every other H is classified far.*

Proof. Each component of a pair within the joint radius is within the individual radius. Since $2t < N\Delta$, individual bounded-distance decoding is unique. Re-encoding and checking the union radius accepts exactly the joint near case. Reed–Solomon bounded-distance decoding, the fixed inverse basis map and the final comparison all take polynomial time. In the far case the prescribed default is deterministic. None of these steps depends on the later circuit or challenge. $\square$

Interpretation. Efficient means polynomial in the oracle length; it does not mean succinct, fast enough for the prover, or recoverable from a public digest. This is an extractor definition, not an extra verifier feature that reveals a holder secret. Its use after Merkle compilation requires the compiler’s extraction theorem. The far branch needs an independent rejection argument, given next.

3 Binding states for a linear opening

For an arbitrary pair H , consider this explicit ideal-oracle test of $\ell(\text{payload}(u)) = a$. The prover sends b , receives fresh coefficients (r_0, r_1) , then sends a complete candidate message $z \in \mathbb{F}^K$. The verifier checks

$$\ell(\text{payload}(z)) = r_0 a + r_1 b$$

and, at q independently uniform indices in the *full* set $[N]$, checks

$$E(z)_j = r_0 H_{0,j} + r_1 H_{1,j}.$$

The coefficient distribution must have the following two stated properties:

$$\begin{aligned} \Pr[r_0 x + r_1 y = 0] &\leq \beta && \text{for every nonzero } (x, y), \\ \Pr[\text{dist}(r_0 H_0 + r_1 H_1, E(\mathbb{F}^K)) < \delta] &\leq \epsilon_{\text{PG}} && \text{if } \text{dist}(H, E(\mathbb{F}^K)^2) \geq \delta. \end{aligned}$$

These are premises about the chosen code and sampler, not measured constants.

Proposition 2 (Linear round-by-round binding). *If $a \neq \ell(d(H))$, the test has a false-claim state function with*

$$b_{\text{lin}} \leq \max\{\beta, \epsilon_{\text{PG}}, (1 - \delta)^q\}.$$

The same rejection reasoning applies to every claimed a when H is far. The entire vector of q query indices is sampled as one verifier move.

Proof. In the near case the recovered messages are fixed before the coefficient challenge. Unless

$$r_0(\ell(\text{payload}(u)) - a) + r_1(\ell(\text{payload}(v)) - b) = 0,$$

which has probability at most β , an explicitly accepted z differs from $r_0u + r_1v$. Distinct codewords differ on at least a Δ fraction, whereas the commitment differs from its decoded pair on less than a δ fraction. Thus the query comparison has at least a $\Delta - \delta > \delta$ mismatch fraction.

In the far case, outside the proximity-generator exceptional event, the combined raw word is at distance at least δ from every codeword, including $E(z)$. Either the explicit equation rejects or all final queries miss at least a δ fraction of bad positions. The latter has probability at most $(1 - \delta)^q$.

Set the state to zero initially. At the coefficient move mark the relevant exceptional event as state one. Otherwise remain zero through the prover's z message, and at the final batch query move mark acceptance as state one. Once one, retain one. A prover move cannot change zero to one, and a fully accepted transcript has state one. The two possible verifier transitions have the stated bounds. This proves a per-round bound, not merely a terminal union bound. $\square$

Query scheduling matters. Sampling the indices one by one as independent protocol rounds would give a different per-round state analysis. The batch above is part of the proposed profile and cannot be silently replaced by an implementation's existing transcript schedule.

4 A query-domain discrepancy and an algebraic counterexample

The retrieved text of VEIL Figure 4, step 8, specifies $[n + k]$, whereas the commitment has codeword length m and the accompanying distance argument uses that length [1]. This should be treated as a possible typographical discrepancy requiring explicit resolution. The candidate in this note specifies queries over the **entire encoded domain**. The following shows why interpreting the smaller range literally is unsafe. It is an original algebraic counterexample to that restricted-domain variant, not an executed native attack or a claim against an implementation using the full domain.

Counterexample 3 (An unchecked polynomial correction). *Let a message be interpolated on $K = n + k$ distinct points A and encoded on a disjoint set B of $N > K$ points. Let the product code contain every polynomial of degree less than $2K$, and let its reduction map return evaluations at the logical points in A . Suppose all product-check queries are restricted to a fixed subset $S \subset B$ of size K . Then a prover can alter a product-check polynomial without changing any queried symbol, while changing its logical residual to any prescribed vector.*

Proof. Let $Z_S(X) = \prod_{b \in S} (X - b)$. Because $A \cap S = \emptyset$, $Z_S(a) \neq 0$ for every $a \in A$. For any desired correction $e \in \mathbb{F}^n$, interpolate a polynomial T of degree less than K with

$$T(a_j) = e_j / Z_S(a_j) \quad (1 \leq j \leq n),$$

choosing the remaining $K - n$ interpolation values arbitrarily. Then $G = Z_S T$ has degree less than $2K$, vanishes on every queried position and has the desired logical correction. Replacing a product polynomial Φ by $\Phi + G$ preserves every restricted query response.

In particular, for a false relation $a \star b \neq c$, write the honestly formed blended product's logical reduction as $s_1(a \star b - c) + s_2r$. Choose $e = -s_1(a \star b - c)$. The altered polynomial reduces to s_2r , so the explicit linear test passes with the honest mask claim. The independent additive combination can be formed honestly. Both tests then pass although the original product relation is false. $\square$

Required correction for this candidate: sample uniformly from all N encoded positions, after the candidate polynomials are fixed. With a nonzero G and full-domain sampling, code distance supplies a detection probability. Merely increasing the number of queries within the same restricted set does not fix this counterexample.

5 An outer round-by-round binding derivation

This section gives a derivation for a specified, corrected code-based outer evaluator. It does not assert that the existing IronSpartan implementation has these states or parameters. Use an efficiently decodable multiplicative Reed–Solomon code and its product code, with relative distances $\Delta_\times, \Delta_\star$. Choose

$$0 < \delta_\star < \frac{1}{2} \min\{\Delta_\times, \Delta_\star\}.$$

Decode interleaved tuples with the decoder of Section 2, using the appropriate code. For a five-column product commitment $(H_a, H_b, H_c, H_+, H_\times)$, analyse

$$(H_a, H_b, H_c, H_+), \quad (H_a H_b - H_c, H_\times).$$

Use the first tuple’s fixed decoded logical triple when it is near, and a fixed default otherwise. When both tuples are near, their decoded product relation agrees: the actual product differs from the product of the decoded first tuple only on its union error positions; two product-code words both that close must be equal by distance.

Let ϵ_4, ϵ_2 bound the two proximity-generator exceptions, β_4, β_2 their linear biases, and β_u the bias of the random linear functional testing the logical product residual. A false product or evaluation claim can survive only through the following events:

1. A nonzero logical product residual is annihilated by that functional: probability at most β_u .
2. At the product-combination challenge, a malformed tuple becomes close to the product code, or a nonzero decoded residual cancels: at most $\epsilon_2 + \beta_2$.
3. At the additive-combination challenge, the analogous event occurs: at most $\epsilon_4 + \beta_4$.
4. A final full-domain query batch misses a mismatch fraction of at least $\delta_\star$: at most $(1 - \delta_\star)^q$.

Proposition 4 (Product-check state bound). *Under these code, sampler, reduction-map and full-domain premises, a conservative round-by-round binding bound is*

$$b_{\text{had}} \leq \max\{\beta_u, \epsilon_2 + \beta_2, \epsilon_4 + \beta_4, (1 - \delta_\star)^q\}.$$

Proof. Use the first applicable false-condition branch, determined by the fixed decoded tuples. For a far tuple, the corresponding proximity challenge supplies the exceptional event; outside it every candidate codeword has the stated query mismatch. For near tuples with a false product, first test annihilation of the nonzero logical residual, then cancellation at the product challenge. For correct products with a false linear claim, use the additive challenge. All scalar mask claims precede the challenge that could cancel their error.

Mark these exceptional verifier transitions as state one, retaining one thereafter. While state is zero, a prover can only send a candidate satisfying the explicit checks with the stated query mismatch, or force rejection. At the final batch, acceptance has the last probability bound. The state cannot become one on a prover move. This also handles malformed oracles: no decoded default is assumed to be their genuine opening. $\square$

Now let the outer circuit be expressed as an R1CS on a *fixed committed variable layout*, with public constants actually enforced. Write its matrices as A, B, C and the decoded committed vector as h . A separate triple oracle is committed before a fresh functional ℓ . It purports to contain $a = Ah, b = Bh, c = Ch$. Its three evaluation claims precede an independent scalar θ , which batches their consistency with h .

Proposition 5 (Adaptive-statement outer binding). *Suppose the code parameters and decoder are fixed before the pad commitment, the circuit matrices are fixed before the outer protocol’s fresh challenges, and all public constant and layout checks are enforced. For a false circuit evaluation at $d(H)$,*

$$b_{\text{CE}} \leq \max\{3\beta_\ell, 2/Q, b_{\text{had}}, b_{\text{lin}}\}$$

is a conservative round-by-round binding bound for the described reduction. It applies when the circuit is chosen after H and after the inner transcript.

Proof. The decoded pad is independent of the subsequent choice of matrices. Once those matrices and the triple oracle are fixed, all three vector discrepancies are fixed. A nonzero discrepancy can have zero ℓ -projection with probability at most β_ℓ ; a union bound covers three discrepancies. If the three claimed values differ from the projections of Ah, Bh, Ch , their batched discrepancy is a nonzero polynomial of degree at most two in θ . Otherwise, outside the projection exceptions, a true product claim and a true linear consistency claim would imply $(Ah) \star (Bh) = Ch$, contradicting the hypothesis.

Consequently at least one component claim is false. Fix an applicable component and use its false-claim state; a far pad or triple commitment uses the corresponding unconditional malformed-oracle branch. Keep the state zero through intervening prover moves and unrelated component messages. Mark projection/batching exceptions at their verifier moves, or use the selected component’s state. Full acceptance forces state one. Fresh outer challenges are independent even after conditioning on the earlier, adaptively selected circuit. $\square$

These propositions depend on explicit proximity/bias bounds and the stated algebraic maps. They provide a derivation strategy and symbolic per-round bounds for this candidate; they do not assign numerical security to native parameters or replace an independent review.

6 Input pads, auxiliary wires and the constant wire

An early commitment to input pads must not be confused with a later commitment to all auxiliary circuit wires. The native symbolic verifier may introduce multiplication wires after challenges and openings are known. Simply declaring those wires to have been in the early commitment is invalid.

One precise candidate amendment is to fix a public maximum layout for the original sensitive messages *and* all multiplication auxiliaries of the opening-verifier computation. Commit enough independent pads for both. After the inner protocol finishes, send the auxiliary values encrypted under their reserved, unused pad entries. These are extra prover messages, not new verifier challenges. The outer relation checks all multiplication definitions, the final decision and $u = c - d(H)$ in that one expanded variable vector. Unused slots have explicit zero constraints. Variable message lengths and secret-dependent layout choices are forbidden.

This amendment preserves knowledge only if the original base acceptance predicate is equivalent to existence of those correctly constrained auxiliaries. It preserves the privacy argument below because the unencrypted expanded vector is computed independently of its new pads. It adds communication and constraints; its cost has not been measured.

Constant-wire condition. The value 1 is a fixed public input. Homogenising the equations and then treating that coordinate as a free committed secret would permit a spurious all-zero solution. The exact outer evaluator must enforce the public-input segment or use an explicitly analysed affine/translated code interface. The preceding state bounds assume this condition; this note does not silently certify a particular native implementation of that interface.

7 Conditional preservation of knowledge

Definition 6 (Required base extraction interface). *Let P_0 be the entire base IOP, including the selected algebraic opening protocol and any auxiliary-message extension. It must have an efficient extractor E_0 and a round-by-round knowledge state S_0 of error k_0 . On a verifier move whose zero-to-one transition probability exceeds k_0 , E_0 takes the preceding transcript and returns a witness satisfying the complete authentication relation. Its oracle interpretation must use the same fixed decoded trace and, where required by the reduction, the same decoded Y for both IntMul claims.*

The definition is stronger than a claim that each standalone check is sound. It is also not established by the phrase “knowledge extraction at each stage” in a specification.

Proposition 7 (Fixed-pad wrapper knowledge, conditional). *Assume the base interface above; a fixed efficient pad decoder d ; the outer false-evaluation state bound b_{CE} for adaptively selected circuits; and an efficient exact prefix translation T replacing each ciphertext c_i by $c_i - d(H)_i$ without changing the base*

challenge distribution or oracle identities. Then the ideal-oracle wrapper has round-by-round knowledge error at most

$$k_{\text{wrap}} = \max\{k_0, b_{\text{CE}}\}.$$

This is a conditional statement, not the numerical security of PQ-DID.

Proof. Before the base protocol starts, the state is zero. During the base phase use the running logical OR of the values of S_0 on all translated prefixes. Taking this monotone closure does not enlarge a zero-to-one transition probability: if the wrapper state is zero, all earlier base states and the current one are zero. A prover move cannot create a one.

For a base verifier move whose transition probability exceeds k_{wrap} , the exact translation gives a base transition exceeding k_0 . Apply E_0 to the translated pre-challenge prefix. The fixed pad is the same throughout, and the base extractor's object-consistency premise supplies one authentication witness and the same committed objects. The extractor does not obtain separate witnesses independently from each opening.

If the state has become one, retain it throughout the outer suffix. Otherwise the completed base state is zero, which implies rejection by its verifier. Therefore the actual circuit evaluated at $d(H)$ is false. Start its outer binding state at zero and use that state through the suffix. Its verifier transitions have probability at most b_{CE} , so none exceeds k_{wrap} and demands a new authentication extractor. Prover moves preserve zero. A fully accepted wrapper transcript cannot have state zero.

The wrapper extractor is $E_0(x, T(\tau))$ at the relevant base transition, and arbitrary elsewhere. Its running time is polynomial in the transcript/oracle lengths by the decoder and translation premises. $\square$

What this resolves. The wrapper does not inherently need a second extractor that somehow reconciles separately extracted hidden terminal slots. Fixed decoding and a false-evaluation state are sufficient for the conditional reduction above.

What remains. The exact native base interface in the definition has not been proved here. In particular, the IntMul/logup reductions, trace packing, heterogeneous domains, opening reductions and query schedule need state/extractor correspondence. Ordinary soundness of those components, or extraction of any satisfying witness without the required object correspondence, is not a substitute.

8 Complete ideal-view simulation and its boundary

For each inner logical oracle a_i , choose a linear embedding $P_i = J_i a_i + B_i \rho_i$ with decoder $D_i J_i = I$ and $D_i B_i = 0$. Fix the complete ideal verifier coin tape. The logical payloads may be correlated, but must be independent of all new embedding coins and companions. Choose independent uniform Ω_i and set

$$V_i = (1 - \gamma)P_i + \gamma\Omega_i, \quad \gamma \neq 0.$$

For each allowed set of distinct disclosed positions I , require that $L_I B_i$ has full row rank, where L_I is restriction of the actual encoding map. Count all scalar positions in a leaf/coset and all openings using that encoding, not merely API query calls.

Proposition 8 (Joint shielded distribution). *Under these premises, the joint distribution across all oracles of*

$$(V_i, L_I P_i, L_I \Omega_i)_i$$

does not depend on the logical payload tuple. Public functions of this view may also be released. Adaptive position choices are permitted only from the already simulated view, public inputs and independent coins, with the rank condition holding on every resulting set.

Proof. For each fixed P_i , the bijection $\Omega_i \mapsto V_i$ makes V_i uniform and independent of P_i . Full row rank makes $L_I P_i$ uniform independently of the payload and of V_i . The companion symbols are determined by

$$L_I \Omega_i = \gamma^{-1}(L_I V_i - (1 - \gamma)L_I P_i).$$

Independent masks give the joint result even for correlated logical payloads. Conditional on already answered independent rows, a new independent row remains uniform; cache repeated positions. This proves the adaptive extension within the stated view. It does not justify queries conditioned on additional witness-dependent data. $\square$

The stronger view containing whole V_i is used only for this distribution argument. It does not assert that native Binius transmits every V_i . A native folding transcript is covered only after showing it is a public function of these vectors and public coins; unclassified scalars must remain encrypted.

For the wrapper composition below, the unencrypted base transcript and shielded view must be generated independently of the wrapper pad. A sufficient schedule fixes base challenges and query choices from the public input and the independent ideal-verifier coin tape. The adaptive-query extension above does not, by itself, justify choosing base queries from wrapper ciphertexts: that could make the shielded view depend on the pad. Such a variant would require a different joint online argument. Fiat–Shamir dependencies are introduced only through the separately matched compiler, not assumed away in a native hash transcript.

Let s be the entire shielded inner view and u the complete expanded sensitive arithmetic transcript. Sample an independent uniform pad h and send $c = u + h$. No pad-oracle opening is permitted before the outer evaluation. Let its exact public input be $X = (x, s, r, c, \text{layout})$ and predicate $C_X(h) = 0$.

Proposition 9 (Conditional complete ideal-IOP simulator). *If the entire s view is simulatable from (x, r) as above, and the selected outer commit-and-prove evaluator is zero-knowledge uniformly for the valid adaptively selected X , with fresh auxiliary commitment randomness, then the following simulates the complete ideal-IOP verifier view: sample s with its simulator; sample c uniformly; invoke the outer simulator on X and the outer verifier’s coins.*

Proof. Because h is independent of (s, u) , $c = u + h$ is uniform and independent of the entire pair. Thus (s, c) has exactly the simulated distribution. Conditional on any fixed (s, c, r) , the honest execution yields some h satisfying the outer predicate. The outer ZK guarantee removes dependence on that witness. Its unused commitment randomness is independent of the inner view conditional on h . Mixing over the possible conditional witnesses preserves the same simulated distribution. Approximate simulators contribute their stated statistical errors. $\square$

The model boundary is essential. An ideal IOP commitment is an oracle string whose contents the verifier sees only through permitted queries. It is not an early concrete hash root. The argument above does not permit erasing already observed native roots from a conditional distribution. It instead establishes the input property required before a matching cryptographic compiler can be applied.

All original targets, individual companion claims, private reduced evaluations and any other witness-dependent arithmetic messages must be inside u . Reused pads, leaked pad openings, mask-derived layout choices or payloads depending on the new companions invalidate the proof unless separately analysed. The native disclosure inventory has not yet established these premises exhaustively.

9 Quantum compilation, exceptional events and finite parameters

Use a specifically matched IOP-to-argument compiler, not an informal assertion that a hash-based backend is post-quantum. The cited QROM theorem separates soundness, knowledge and zero-knowledge hypotheses and has a specified hash-chain construction [2].

The proposed application requires all of the following simultaneously:

1. The preceding IOP’s knowledge and complete-view simulation premises are established for the whole relation.
2. Salted commitments, message/oracle serialization, challenge order and the statement/layout/context binding match the selected compiler argument. A secret salt is not a replacement for hiding an exposed field element.
3. Every use of the public hash, including adaptive queries, is counted; no concrete SHA implementation is treated as a programmable oracle by assertion.
4. Field, code distance, query count, salt and digest sizes are selected with the reduction losses, not just the component signature category.

For orientation, the cited knowledge reduction contains terms of the form

$$\Pr[\text{extract}] \geq \Omega\left(\mu - t^2 k_{\text{wrap}} - \frac{t^3}{2^\lambda}\right),$$

with its stated query-overhead and model conditions [2]. This is not a computed bound for the current project. A field of size 2^{128} does not by itself establish 128-bit overall quantum security after those losses.

The profile must specify exceptional challenges. For example, for the affine blend used above, $\gamma = 0$ defeats the uniformity argument and $\gamma = 1$ removes the original oracle’s coefficient. One explicit candidate rule is immediate rejection on either value, without resampling. Its honest rejection probability is at most $2/Q$ per such uniform challenge. Other bad coefficients, including outer masking degeneracies, require their own branches and count. This is a proposed rule, not a change to the active profile.

No quantum/native conclusion follows yet. The mathematical interfaces in this note do not certify the retained Binius compiler, its hash suite, current parameters, or the complete native reduction. The signing/sampler-tail and concrete-hash obligations in the project also remain separate.

10 Exact remaining construction and implementation gates

The following are implementation requirements, not permission to execute a new package.

Gate	Required evidence
BASE-RBR	A prefix state/extractor map for the chosen <i>complete</i> base protocol: trace, both same- Y openings, all IntMul/logup reductions, packing, range checks, and opening verification. Give degrees, exceptional events and a single decoded credential witness.
CE-EXACT	Instantiate the outer code and reduction maps, full query domain, constant/public-input enforcement, auxiliary-message layout and challenge samplers used by Sections 2–7. Establish proximity/bias parameters and native correspondence.
VIEW-ALL	Classify every native output, including roots, claims, final folded buffers, metadata and failure behaviour. Show every unencrypted inner value is a function of the permitted shielded view. Preserve fresh-mask independence and aggregate query-rank bounds.
COMPILER	Match the complete IOP to the specific salted commitment/Fiat-Shamir theorem, including statement binding and adaptive query accounting. Do not substitute a classical result for a quantum knowledge result.
RESOURCE	Count simultaneous witness, original/companion encodings, outer matrices, auxiliary pads, Merkle buffers and output storage before any allocation/proof. No complete-authentication performance claim is established here.

Native correctness obligations already known. Both IntMul openings must use the same Y identity. The equality operand uses the actual returned `pushforward_eval_point`, derived from the last `table.n_vars` coordinates of `node_point`. Both it and the power-table operand must lift through the same embedding decoder. Authenticated leaf values used outside the outer proof must be the identical public values used by its verification circuit. No unauthenticated duplicates may substitute for them.

Recommended next deliverable. Supply BASE-RBR for this named candidate, with CE-EXACT and VIEW-ALL discrepancies recorded against the explicit construction above. Do not repeat a general backend survey or the completed embedding tests. If the necessary state/extractor map cannot be supplied, return that concrete failure instead of declaring the candidate admitted. A separately authorised synthetic implementation could test a specified component or source correspondence while research remains open, but that would not admit a complete private-authentication profile. Any proof-producing pilot also needs its explicit resource and execution authorisation; it cannot spend an unused corrective slot.

Manuscript. Keep this note separate from the manuscript. An adopted replacement would require updating Sections VII–VIII and the affected interfaces/assumptions in II–VI. Neither the old BC-1 profile nor the current research note describes a measured complete private-authentication implementation.

Evidence and limits of this deliverable

This work consists of source reading and written algebraic derivations. No empirical security test, native forgery, formal proof-assistant verification or independent cryptographic audit is claimed. The counterexample concerns a specified restricted-query variant. The new lemmas are offered for technical review with all of their premises; they do not close CV-ONLINE-SAME-OBJECT-1 for the old construction or retrospectively change its outcome.

No WSL allocation, test count, build allowance, benchmark result or proof-attempt ledger was consumed or amended by this document. Production verification remains fail-closed and proving/isolation remain paused. Completion by 31 October 2026 is not established by these derivations.

References

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- [3] Retained project evidence: oct31_binius64_replacement_decision.md; oct31_binius64_g0.md; stage3_binius_embedding_correspondence.md; stage3_binius_committed_view_correspondence.md; and PQ_DID_Wrapper_Repair.zip. Native source identity is recorded in Section 1.